

Causal Sliding-Window Prediction of Imminent Obstructive Sleep Apnea from Single-Channel PPG using 1D-CNN

Abstract—Obstructive Sleep Apnea (OSA) continues to be underdiagnosed because of the high cost and complexity of polysomnography (PSG), thereby creating a demand for more affordable and accessible screening solutions. Photoplethysmography (PPG) provides a continuous, non-invasive alternative; however, accurate short-term forecasting of apneic episodes using single-channel PPG alone is still a difficult task. In this letter, we introduce an automated deep learning solution for ultra-short-term apnea forecasting from raw PPG signals using a customized one-dimensional convolutional neural network (1D-CNN). The solution analyzes 30-second observation segments to forecast the occurrence of apnea or hypopnea episodes in a future 10-second prediction window with a 5-second lead time, maintaining strict temporal causality. The evaluation was conducted on a strictly patient-independent split of the publicly available OSASUD dataset to avoid any potential data leakage. On an unseen hold-out test set of six patients (8,638 windows), the proposed solution attained a sensitivity of 86.03%, a precision of 53.01%, F1-score of 0.656, and AUC of 0.71. The relatively low precision reflects a deliberate sensitivity-prioritized design, appropriate for a clinical screening context where missed detections carry greater risk than false alarms.

Index Terms—Obstructive Sleep Apnea, Photoplethysmography (PPG), Deep Learning, 1D-CNN, Continuous Monitoring

I. INTRODUCTION

OBSTRUCTIVE Sleep Apnea (OSA) is a chronic and pervasive sleep-disordered breathing condition that is estimated to affect one billion adults worldwide [1], [2]. Sleep Apnea can be defined as the recurrent partial or complete obstruction of the upper airway during sleep, which results in the recurrent episodes of breathing pauses. These breathing pauses result in a series of pathological responses, including intermittent hypoxia, sleep fragmentation, and activation of the sympathetic nervous system, which are major risk factors for serious cardiovascular diseases like hypertension, stroke, and heart failure [3]–[5].

Notwithstanding the serious medical consequences, the vast majority of OSA cases remain undiagnosed because of the shortcomings of the existing paradigms for diagnosis. The present gold standard for the diagnosis of OSA is in-lab Polysomnography (PSG) [6]. Although it is highly accurate, PSG is a multi-parametric procedure that requires patients to be hooked up to a large number of sensors—including electroencephalography (EEG), electrocardiography (ECG), electromyography (EMG), respiratory effort belts, and pulse oximetry (SpO₂). As a result, PSG is costly, requires trained technical staff, is uncomfortable for patients and disturbs their normal sleep patterns, and is not suited for screening or home monitoring [7], [8]

Photoplethysmography (PPG) has emerged as a promising non-invasive alternative for continuous physiological monitoring, capable of capturing autonomic nervous system responses to breathing disruptions from a single point of skin contact [9]. Recent breakthroughs in deep learning have proven extremely promising in automating this detection process, with Convolutional Neural Networks (CNNs) demonstrating strong capability in extracting predictive features directly from raw physiological signals [10], eliminating the need for manual feature engineering. Despite this progress, a critical gap remains. Prior PPG-based approaches predominantly rely on post-event classification or multi-modal fusion [11], and strictly causal forward prediction of imminent apneic events from raw single-channel PPG alone remains heavily underexplored. To bridge this gap, this study proposes an automated, strictly causal sliding-window forward prediction framework for OSA using solely raw single-channel PPG signals, notably, without any auxiliary modalities, handcrafted features, or post-hoc analysis. The following sections detail the proposed methodology, experimental setup, and results on a publicly available PPG dataset.

II. MATERIALS AND METHODS

A. Dataset and Preprocessing

We leveraged the publicly available Obstructive Sleep Apnea Stroke Unit Dataset [12]. The dataset provides full-night polysomnographic recordings from 30 patients (ages 27-68 years, BMI 20-42 kg/m²), with diverse severities of obstructive sleep apnea (Apnea-Hypopnea Index: 0.3-96.2 events/hour). The recordings are synchronized PPG, SpO₂, heart rate, and expert-annotated apnea/hypopnea events. Recordings lasted between 7.0 and 11.9 hours per patient.

For this work, we selected only the PPG signal sampled at 100 Hz, excluding all other auxiliary signals to test the performance in a single-modality setup. The raw PPG signals were bandpass filtered (0.5–8.0 Hz) to retain cardiac pulses, z-score normalized to account for inter-subject physiological variability, and motion artifacts exceeding ± 5 SD from the mean were removed.

B. Sliding Window Segmentation

To effectively capture the physiological antecedents of apneic events, we used overlapping sliding window strategy. Each observation window was 30 seconds long (3,000 samples at 100 Hz), sampled with a 1-second stride, resulting in a

96.7% overlap between successive windows. Given an observation window ending at time t , a binary class label was assigned according to the presence of annotated apnea/hypopnea events within a prediction window of $[t + 5s, t + 15s]$ – that is, a 10-second window starting 5 seconds after the end of the observation window. A window was labeled as apnea (Class 1) if any part of an annotated apneic event occurred within this prediction window; otherwise, it was labeled as normal (Class 0). This labeling strategy is intended to leverage the anticipatory cardiovascular patterns that are discernible within the PPG signal, such as amplitude modulation and heart rate variability, which precede actual apneic events by several seconds. The observation window and prediction window were strictly non-overlapping, ensuring no future information leakage. This form of predictive windowing is also practically useful, as it offers a 5-second lead time for intervention or alerting within a real-time monitoring application, and a 10-second prediction window that is broad enough to include variable-onset apneic events without introducing too much label noise from future windows. The observation window does not overlap with the SAME apnea event being predicted.

C. Train-Validation-Test Split Strategy

To guarantee generalization and, to avoid any form of data leakage, we adopted a strict patient-independent data splitting approach. In the context of physiological signal processing, it is important to note that the inclusion of overlapping windows from the same patient in both the training and testing data subsets would artificially inflate model performance, since the neural network would simply learn patient-specific cardiovascular patterns instead of apneic patterns. Consequently, all windows extracted from any given subject were strictly limited to only one subset. To avoid data leakage, a strict patient-independent split was adopted. 20% of patients (6 patients, 1,432 apnea and 7,307 normal windows) were allocated to an entirely unseen hold-out test set, with all remaining windows confined to training. This approach ensures that the final performance metrics calculated on the hold-out test set are a true representation of the actual diagnostic performance of the model on unseen patients in a home-monitoring scenario.

D. Class Imbalance Handling

The polysomnographic data is inherently highly imbalanced, with the number of normal breathing episodes far outnumbering apneic episodes [13]. In our preprocessed dataset, the ratio of apnea to normal windows is roughly 1:5, with apneic episodes making up about 16% of the data. A standard neural network trained on such an imbalanced dataset will naturally learn to predict the majority class, achieving high overall accuracy but at the cost of unacceptably low sensitivity for the minority class.

To mitigate this imbalance without artificially altering the inherent temporal distribution of the physiological data through resampling (such as SMOTE or undersampling), we turned to cost-sensitive learning. We assigned inverse frequency class weights to scale the loss gradient during training:

$$w_c = \frac{N}{K \cdot n_c}, \quad (1)$$

where N is the total number of training samples, $K = 2$ is the number of classes, and n_c is the number of samples in class c . By weighting the objective function with these class weights, the network is encouraged to heavily penalize misclassifications of the minority (apnea) class. This directly addresses the high sensitivity requirement for a screening tool, ensuring that potentially life-threatening apneic episodes are not missed.

E. 1D-CNN Architecture

Although recurrent neural networks (RNNs) and Transformers are popular choices for time series forecasting tasks, we chose to focus on a one-dimensional Convolutional Neural Network (1D-CNN) architecture [14]. 1D-CNNs have much faster training times, fewer trainable parameters, and are generally much more computationally efficient than traditional sequential architectures. Moreover, apneic episodes are characterized by particular localized morphological features in the PPG signal (such as abrupt amplitude changes) [15], which are well-suited to the translation-equivariant properties of convolutional filters.

The architecture is designed to balance feature extraction depth with computational efficiency, prioritizing localized morphological pattern recognition over sequential modeling. The proposed network comprises four convolutional blocks followed by fully-connected classification layers (Fig. 1), intended to extract multi-scale temporal features from raw PPG windows of size 3000×1 .

The initial convolutional layer uses 64 filters with a kernel size of 7 and ReLU activation functions, accompanied by batch normalization [16] to normalize the training process and minimize internal covariate shift. A max pooling layer with a pool size of 2 is used to decrease spatial dimensions, accompanied by dropout regularization [17] with a dropout rate of 0.3 to prevent overfitting. The second convolutional layer doubles the filter depth to 128 with a kernel size of 5, using the same normalization and dropout strategy. Likewise, the third convolutional layer uses 256 filters with a kernel size of 3, accompanied by the same normalization and dropout strategy, but with an increased dropout rate of 0.4 to offer more regularization as the network becomes deeper. The fourth and final convolutional layer uses 512 filters with a kernel size of 3, accompanied by global average pooling to collapse spatial dimensions into a single value per channel, and dropout with a rate of 0.5 for aggressive regularization.

After the convolutional layers, two dense layers are used to process the learned abstract features. The first dense layer has 128 units with ReLU activation function and a dropout rate of 0.5, while the second dense layer has 64 units with the same activation function. The output layer has a single dense unit with a sigmoid activation function to produce the final binary classification probability.

The progressive filter depth growth from $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ allows the network to learn increasingly abstract representations, from which basic pulse morphology is represented in early layers and complex apnea patterns are represented in deeper layers. On the other hand, the progressively

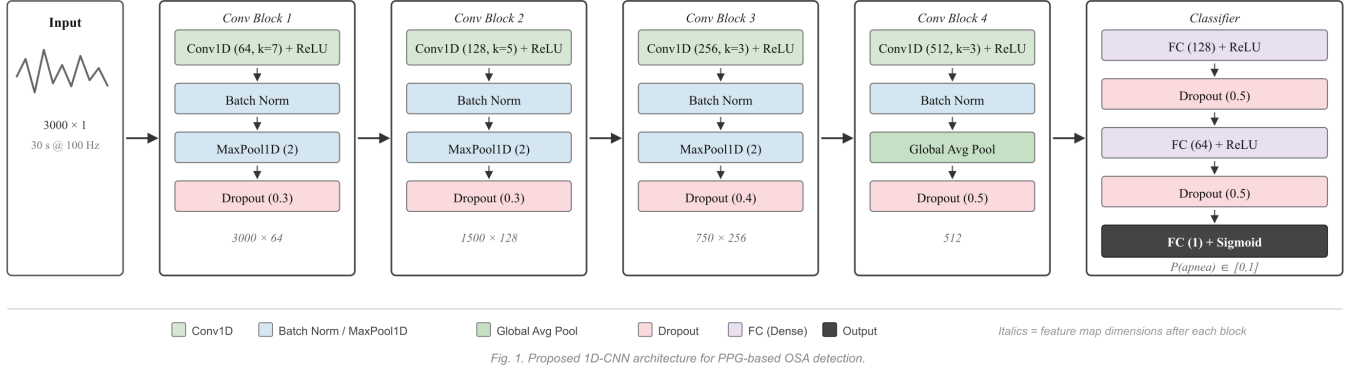


Fig. 1. Proposed 1D-CNN architecture for PPG-based apnea prediction. The network progressively extracts hierarchical features through four convolutional blocks.

smaller kernel sizes from $7 \rightarrow 5 \rightarrow 3 \rightarrow 3$ strike a balance between the receptive field and computational complexity, allowing the network to capture both coarse and fine temporal patterns. Carefully placed batch normalization and dropout layers prevent overfitting to patient-specific baselines.

F. Training Procedure

The 1D-CNN was trained using the Adaptive Moment Estimation (Adam) optimizer with an initial learning rate of $\alpha = 0.001$. To account for the highly skewed distribution of normal versus apneic windows, we optimized the network using a weighted binary cross-entropy loss function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (2)$$

where $y_i \in \{0, 1\}$ represents the true ground-truth label, $\hat{y}_i$ is the predicted probability output by the sigmoid activation, and w_{y_i} is the computed class weight for the respective instance.

The training was performed for a maximum of 30 epochs with a batch size of 64. To avoid overfitting and ensure the best convergence of the weights, an early stopping callback was used, which monitored the validation loss with a patience of 5 epochs. After the validation of the hyperparameters during the 5-fold cross-validation process, the final classification model was trained on the 24 patients. During this final training process, a patient-independent set was internally held out from the 24 patients only for monitoring the validation loss during the early stopping process to avoid overfitting before the model was evaluated on the completely unseen 6-patient hold-out test set.

G. Evaluation Metrics

Performance was evaluated using Sensitivity, Precision, Specificity, F1-Score, and AUC. Sensitivity was prioritized as the primary metric, reflecting the clinical requirement of minimizing missed apneic events [13].

III. RESULTS

A. Cross-Validation Performance

Five-fold GroupKFold cross-validation on the 24-patient training data showed stable results across folds (Table I). The mean cross-validation metrics were $\text{AUC} = 0.7216 \pm 0.0603$, $\text{F1-score} = 0.6927 \pm 0.0156$, $\text{Sensitivity} = 0.8658 \pm 0.0360$, and $\text{Precision} = 0.5782 \pm 0.0209$. The low standard deviations across folds indicate stable generalization, suggesting the model learns apnea-specific cardiovascular patterns rather than overfitting to patient-specific baselines. Five-fold GroupKFold cross-validation was employed on the 24-patient training data, ensuring all windows from a given patient remain in a single fold, preserving patient-independence during hyperparameter validation.

TABLE I
FIVE-FOLD CROSS-VALIDATION RESULTS ON TRAINING SET (24 PATIENTS)

Fold	AUC	F1-Score	Sensitivity	Precision
1	0.7760	0.6964	0.8506	0.5895
2	0.6300	0.6670	0.8793	0.5373
3	0.7530	0.6993	0.8509	0.5935
4	0.7783	0.6865	0.8205	0.5901
5	0.6705	0.7141	0.9274	0.5805
Mean	0.7216	0.6927	0.8658	0.5782
Std	0.0603	0.0156	0.0360	0.0209

B. Hold-Out Test Set Performance

The critical evaluation on 6 completely unseen patients (1,331 apnea windows, 7,307 normal windows) showed clinically viable screening performance. As can be seen in Table II, the model reached a sensitivity of 86.03% (correctly identifying 1,145 out of 1,331 apnea windows) and a precision of 53.01% (1,145 correct positive predictions out of 2,160 total positive predictions). The overall F1-Score was 0.6560, with an AUC of 0.7100. Moreover, the model kept a high specificity of 86.11% (correctly identifying 6,292 out of 7,307 normal windows), which led to a relatively low false positive rate of 13.89%.

TABLE II
PER-PATIENT PERFORMANCE ON HOLD-OUT TEST SET

Patient	Events	Sens.	Prec.	F1	AUC
7	200	0.1389	0.6883	0.2312	0.7380
17	320	0.8531	0.4919	0.6240	0.6623
18	180	0.8630	0.5338	0.6639	0.7097
23	210	0.8627	0.5333	0.6631	0.7046
25	241	0.8465	0.5411	0.6602	0.6736
30	180	0.8444	0.4984	0.6268	0.6988
Overall	1331	0.8603	0.5301	0.6560	0.7100

On the 8,638 test windows, the model produced 1,145 true positives, 186 false negatives, 6,292 true negatives, and 1,015 false positives.

C. Per-Patient Analysis

However, individual patient performance was not identical (Table II), which can be attributed to the variability of apnea severity, quality of PPG signals, and individual physiological differences. Patients 17, 18, 23, 25, and 30 had very consistent prediction performance (sensitivity ranging from 84.4% to 86.3%), whereas Patient 7 had much lower sensitivity.

D. Prediction Examples: Successes and Failures

Fig. 2 shows six temporal predictions for Patient 17, including both successful predictions (examples 2, 3, 4, 5, and 6) and partial misses (example 1). The temporal arrangement for each example includes: a 30-second observation window (green), a 5-second lead time (yellow), and a 10-second prediction window (red) where the apneic event can potentially occur.

Successful predictions (examples 2, 3, 4, 5, and 6) are characterized by observable physiological precursors within the 30-second observation window — most notably, a progressive reduction in PPG pulse amplitude and increased beat-to-beat interval variability in the seconds leading up to the prediction window, consistent with the autonomic nervous system response to impending airway obstruction. These patterns provide the model with sufficient morphological evidence to correctly forecast the apneic event 5–15 seconds ahead. In example 1, while the model correctly flagged apneic activity, additional apnea onset markers fall outside the prediction window, indicating rapid or closely-spaced events where precursor patterns are absent or too subtle to distinguish from normal breathing variability within a single-channel PPG signal alone — highlighting an inherent limitation of single-modality prediction.

In all successful cases, the apnea onset marker (dashed line) falls within the red prediction window, confirming the model’s ability to anticipate the event within the defined lead time. In failed cases, the absence of visible amplitude attenuation or rhythm irregularity in the green observation window explains the model’s inability to forecast the impending event.

IV. DISCUSSION

A. Principal Findings

This work proves the applicability of using a 1D-CNN to predict Obstructive Sleep Apnea events solely from single-

channel PPG signals. With our model having been able to reach a sensitivity of 86.03% and specificity of 86.11% on a completely unseen test set, our model has set a strong foundation for continuous physiological signal monitoring. Most importantly, our approach ensures that the data leakage problem, commonly witnessed in physiological time-series analysis [18], [19], is strictly avoided. This ensures that the apnea-specific cardiovascular patterns were learned by the network and not the patient-specific baselines. The observed precision of 53.01% is an expected consequence of several compounding factors. The dataset is inherently imbalanced (approximately 1:5 apnea to normal windows), and cost-sensitive training deliberately prioritizes sensitivity over precision, which is the clinically appropriate tradeoff for a screening tool where failing to detect a true apneic event carries significantly greater clinical risk than incorrectly flagging a normal breathing episode. Furthermore, our evaluation is strictly more challenging than most comparable works, employing single-channel PPG, causal forward prediction, and a fully patient-independent split, whereas prior works reporting higher precision typically rely on multi-modal inputs or patient-dependent evaluation protocols.

B. Comparison with State-of-the-Art Approaches

The conventional approach to diagnosing OSA is highly dependent on full overnight polysomnography (PSG), which is resource-intensive and uncomfortable for patients [6]. Compared to the latest deep learning baselines, our single-channel PPG model provides highly competitive performance while relying solely on a single physiological input modality. Although Wang *et al.* [10] proved the strong capability of deep CNNs in automatic feature learning for sleep apnea prediction, these state-of-the-art models typically require clinical-grade or multi-channel inputs.

Similarly, the comprehensive benchmark by Chi *et al.* [11] showed that current deep learning models (DANet and GATE) demonstrate binary prediction accuracies between 46% and 87.18% when using comprehensive multi-dimensional feature sets. Our proposed 1D-CNN model demonstrates a similar sensitivity of 86.03% and specificity of 86.11% on strictly single-channel PPG inputs. Unlike complex recurrent networks or transformers, our 1D-CNN achieves highly competitive screening performance with a minimal parameter set, matching the performance thresholds reported by Chi *et al.* [11] using only a single-channel PPG input.

C. Clinical Implications

In light of the extreme global burden of OSA that remains undiagnosed [20] and the direct causal relationship of OSA to cardiovascular disease and hypertension [3], there is a pressing clinical need for the development of widely available, long-term screening modalities [20]. The computational efficiency and relatively small parameter count of the proposed 1D-CNN suggest potential suitability for continuous long-term monitoring scenarios, supporting longitudinal apnea burden analysis without dependence on clinical infrastructure.

Patient 17: CORRECT Pre-Apnea Predictions (True Positives: 285)

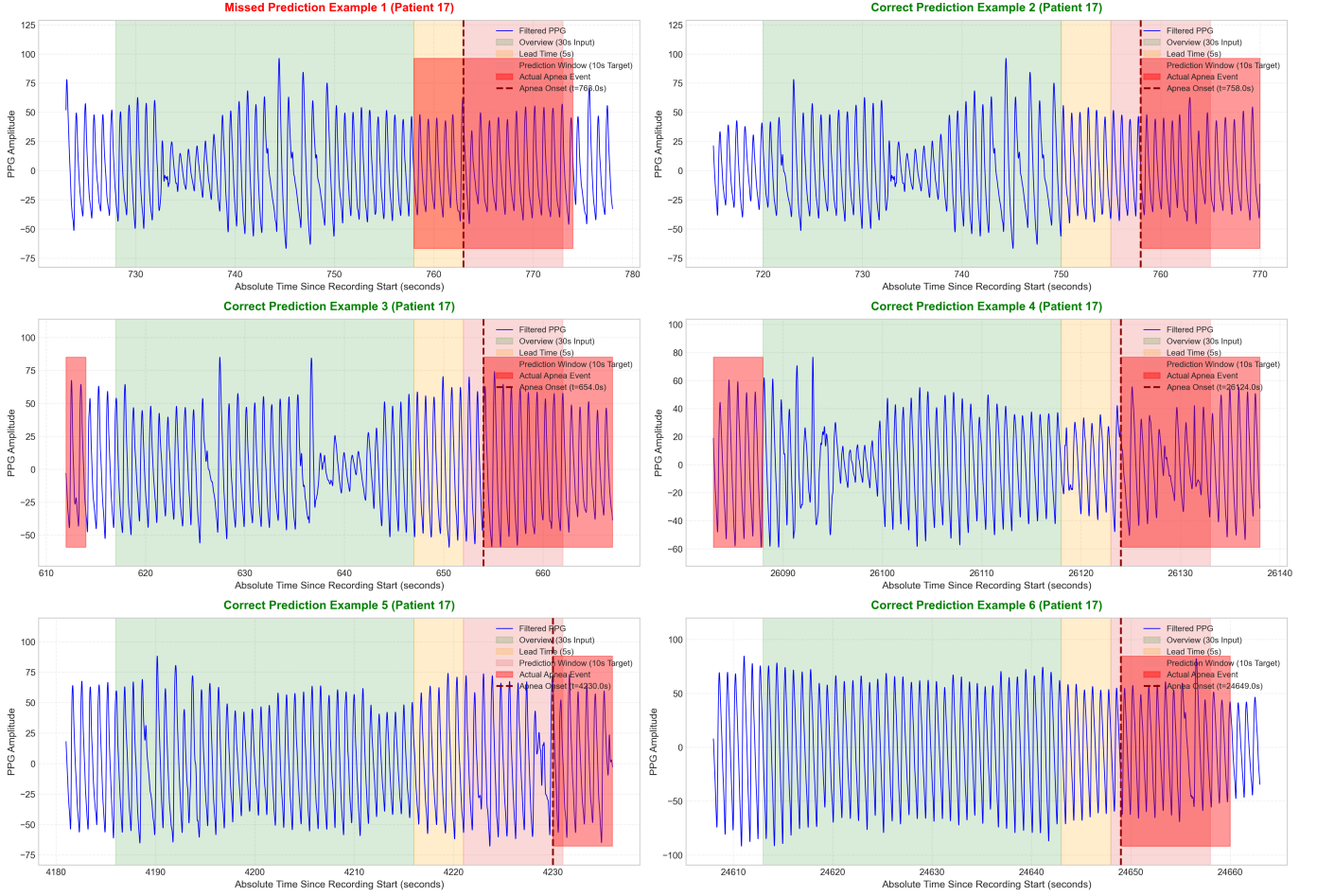


Fig. 2. Prediction examples for Patient 17: Correct predictions (examples 2, 3, 4, 5, 6) where the model successfully predicts apnea 5–15 seconds ahead, and partial misses (example 1) where apnea events (dashed lines) occur unpredicted. Green: observation window; Yellow: lead time; Red: prediction window. This illustrates the model’s 85.3% sensitivity for this specific patient, correctly anticipating most events while missing rapid-onset cases.

D. Limitations and Future Directions

Despite its robust screening performance, the proposed approach also has some significant shortcomings. The model has a 13.89% false positive rate, which, if implemented as a real-time notification system, may potentially cause alarm fatigue. Moreover, as seen in the per-patient analysis (Table II) and the particular prediction failures (Fig. 2), the model sometimes has difficulty identifying rapid-onset or very subtle apneic episodes that fail to trigger immediate, strong cardiovascular precursors in the PPG signal.

Future research will be driven by two main avenues. Firstly, the addition of a complementary modality such as single-lead ECG, which captures cardiac electrical precursors to airway obstruction, may prove effective at lowering the false positive rate by providing additional anticipatory physiological information. Secondly, the addition of a lightweight temporal convolutional network (TCN) [21] appended to the end of the feature extraction blocks may prove to be effective at allowing the network to learn longer-term dependencies over a period of several minutes of sleep without incurring a large

computational cost.

V. CONCLUSION

In this study, we have proposed an automated and deep learning solution for the prediction of imminent OSA events solely from raw single-channel PPG signals. Through the design of a lightweight one-dimensional Convolutional Neural Network (1D-CNN) and by strictly following a patient-independent evaluation protocol, we have shown that the peripheral blood volume pulse is rich enough in morphological features for apnea prediction. On a test set of completely unseen patients, our solution has shown a good diagnostic balance with a sensitivity of 86.03% and specificity of 86.11%.

Crucially, this method effectively addresses the ubiquitous problem of data leakage in physiological machine learning and obviates the need for complex multi-sensor medical infrastructure. The computational efficiency and small parameter count of the proposed 1D-CNN suggest potential suitability for resource-constrained platforms, which may support scalable and non-invasive continuous monitoring. Future work will

focus on the incorporation of complementary anticipatory modalities such as single-lead ECG, alongside sophisticated temporal modeling to further reduce false positive rates, ultimately improving the accessibility of early OSA screening.

REFERENCES

- [1] A. V. Benjafield, N. T. Ayas, P. R. Eastwood *et al.*, “Estimation of the global prevalence and burden of obstructive sleep apnoea: a literature-based analysis,” *The Lancet Respiratory Medicine*, vol. 7, no. 8, pp. 687–698, 2019.
- [2] C. V. Senaratna, J. L. Perret, C. J. Lodge *et al.*, “Prevalence of obstructive sleep apnea in the general population: a systematic review,” *Sleep medicine reviews*, vol. 34, pp. 70–81, 2017.
- [3] V. K. Somers, D. P. White, R. Amin *et al.*, “Sleep apnea and cardiovascular disease: an American Heart Association/American College of Cardiology Foundation scientific statement,” *Circulation*, vol. 118, no. 10, pp. 1080–1111, 2008.
- [4] J. M. Marin, S. J. Carrizo, E. Vicente, and A. G. Agusti, “Long-term cardiovascular outcomes in men with obstructive sleep apnoea-hypopnoea,” *The Lancet*, vol. 365, no. 9464, pp. 1046–1053, 2005.
- [5] H. K. Yaggi, J. Concato, W. N. Kernan *et al.*, “Obstructive sleep apnea as a risk factor for stroke and death,” *New England Journal of Medicine*, vol. 353, no. 19, pp. 2034–2041, 2005.
- [6] V. K. Kapur, D. H. Auckley, S. Chowdhuri *et al.*, “Clinical practice guideline for diagnostic testing for adult obstructive sleep apnea,” *Journal of Clinical Sleep Medicine*, vol. 13, no. 3, pp. 479–504, 2017.
- [7] W. W. Flemons, M. R. Littner, J. A. Rowley *et al.*, “Home diagnosis of sleep apnea: a systematic review of the literature,” *Chest*, vol. 124, no. 4, pp. 1543–1579, 2003.
- [8] N. A. Collop, W. M. Anderson, B. Boehlecke *et al.*, “Clinical guidelines for the use of unattended portable monitors in the diagnosis of obstructive sleep apnea in adult patients,” *Journal of Clinical Sleep Medicine*, vol. 3, no. 7, pp. 737–747, 2007.
- [9] J. Allen, “Photoplethysmography and its application in clinical physiological measurement,” *Physiological Measurement*, vol. 28, no. 3, p. R1, 2007.
- [10] E. Wang, I. Koprinska, and B. Jeffries, “Sleep apnea prediction using deep learning,” *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 11, pp. 5644–5654, 2023.
- [11] H. T. K. Chi, A. Dabbs-Brown, A. Jurek-Loughrey *et al.*, “Obstructive sleep apnea prediction: a comprehensive review and comparative study,” *Machine Learning*, vol. 115, no. 2, p. 29, 2026.
- [12] A. Bernardini, A. Brunello, G. L. Gigli, A. Montanari, and N. Saccomanno, “OSASUD: A dataset of stroke unit recordings for the detection of obstructive sleep apnea syndrome,” *Scientific Data*, vol. 9, no. 1, p. 177, 2022.
- [13] Y. Sun, A. K. Wong, and M. S. Kamel, “Classification of imbalanced data: a review,” *International Journal of Pattern Recognition and Artificial Intelligence*, vol. 23, no. 04, pp. 687–719, 2009.
- [14] S. Kiranyaz, O. Avci, O. Abdeljaber *et al.*, “1D convolutional neural networks and applications: a survey,” *Mechanical Systems and Signal Processing*, vol. 151, p. 107398, 2021.
- [15] E. Gil, M. Mendez, J. M. Vergara *et al.*, “Discrimination of sleep-apnea-related decreases in the amplitude fluctuations of PPG signal in children by HRV analysis,” *IEEE Transactions on Biomedical Engineering*, vol. 56, no. 4, pp. 1005–1014, 2009.
- [16] S. Ioffe and C. Szegedy, “Batch normalization: accelerating deep network training by reducing internal covariate shift,” in *International Conference on Machine Learning*. PMLR, 2015, pp. 448–456.
- [17] N. Srivastava, G. Hinton, A. Krizhevsky *et al.*, “Dropout: a simple way to prevent neural networks from overfitting,” *Journal of Machine Learning Research*, vol. 15, no. 1, pp. 1929–1958, 2014.
- [18] S. Saeb, L. Lonini, A. Jayaraman *et al.*, “The need to report machine learning performance with participant-level cross-validation,” *GigaScience*, vol. 6, no. 5, p. gix019, 2017.
- [19] G. Vandewiele, I. Dehaene, G. Kovács *et al.*, “Overly optimistic prediction results on imbalanced data: a case study of flaws and benefits when applying over-sampling,” *Artificial Intelligence in Medicine*, vol. 111, p. 101987, 2021.
- [20] S. R. Patel, “The global burden of obstructive sleep apnea,” *American Journal of Respiratory and Critical Care Medicine*, vol. 199, no. 12, pp. 1442–1444, 2019.
- [21] S. Bai, J. Z. Kolter, and V. Koltun, “An empirical evaluation of generic convolutional and recurrent networks for sequence modeling,” *arXiv preprint arXiv:1803.01271*, 2018.